Tejasva Singh Chouhan

Indore, India | +91 62616 60095 | tejasva12112004@gmail.com
linkedin.com/in/tejasva-singh-chouhan-859bab31a | github.com/tejasvasingh2004 | leetcode.com/u/tejasvasingh44

Education

Shri G. S. Institute of Technology and Science (SGSITS) 2024 – 2028
B.Tech in Computer Engineering — ***CGPA: 8.17/10*** *Indore, India*

Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

St. Joseph's Convent Senior Secondary School 2023
CBSE Class XII: ***90.2%*** | *Class X:* ***93.8%*** *Sagar, India*

Experience

Maulana Azad National Institute of Technology (MANIT) May 2026 – Jul 2026
Research Intern – BharatGen Vision-Language Models *Bhopal, India*

- Engineered multilingual data pipelines combining OCR, image-text preprocessing, and language-specific tokenization for **5,000+ image-text samples** across **21 Indian languages**.
- Ran **20+ experiments** across **4 preprocessing/model configurations**, analyzing output quality and error patterns to guide multimodal VLM workflow design.

InDhan Pay Pvt. Ltd. 2025
Backend & AI Intern | Node.js, PostgreSQL, Knex.js, AWS SNS, Joi *Remote*

- Designed reusable **AWS SNS notification services** for **15+ event types** across employee lifecycle, dealership, credit-account, and sales/payment flows, with role-based and individual-user routing.
- Built a reusable **Joi validation framework** and integrated request-validation middleware across **40+ endpoints** in **5 API modules** (auth, employee, dealership, sales, credit).
- Developed credit-customer APIs with phone-number masking, role-based filtering, and safer response handling; evolved PostgreSQL schemas through **10+ Knex migrations** and transactional queries.
- Fixed **20+ production bugs**, refactored duplicated logic, and improved error handling and Swagger/OpenAPI documentation across REST services.

Projects

NeuroFlow – AI Workspace & Coding Agent Platform | *TypeScript, React, Node.js, PostgreSQL, Redis, BullMQ, Docker*

- Built a full-stack AI agent platform spanning **13 packages**, ∼**24K lines of TypeScript**, and **103 REST/SSE endpoints**, routing requests across **6 execution strategies** and multiple LLM providers.
- Engineered BullMQ workers with retries, provider fallback, cancellation, and crash recovery, achieving **20–70 ms queue latency** and **0.4–0.8 s time-to-first-token**.
- Implemented a coding agent with **27 validated tools** running in isolated Docker sandboxes, with human-approved diffs, **AES-256-GCM** credential encryption, and branch isolation.
- Wrote **194 automated tests** (incl. 13 Playwright E2E) and shipped real-time IDE features for code editing, diff review, terminal access, and run replay.

Mini Database Engine – B+ Tree Relational Database | *Java 17, Spring Boot, JavaScript, HTML*

- Built a relational database engine from scratch (∼**3.3K lines**) on an **order-50 B+ Tree** with $O(\log n)$ lookups, inserts, and deletes, including automatic node splitting and merging.
- Designed a **4-layer architecture** (storage, schema management, query/transaction processing, REST API) with JSON persistence and snapshot-based rollback for **atomic transactions**.
- Exposed **11 REST endpoints** supporting **6 data types**, plus a web UI for CRUD, conditional queries, pagination, and multi-step transaction building.

Technical Skills

Languages: C, C++, Java, JavaScript, TypeScript, SQL
Backend: Node.js, Express.js, Fastify, Spring Boot, REST APIs, SSE, BullMQ, JWT
Frontend: React, HTML, CSS, Tailwind CSS
Databases: PostgreSQL, Redis, Knex.js, Schema Design, Indexing (B+ Trees), Transactions
AI/ML: LLM Integration, AI Agents, Vision-Language Models, OCR, Multilingual Data Pipelines
Cloud & Tools: AWS (SNS), Docker, Git, GitHub, Playwright, OpenTelemetry, Swagger/OpenAPI, Joi

Achievements

- **Amazon ML Challenge 2026:** All-India Rank **507** among **34,000+ teams** (**top 1.5%**), with an evaluation score of **98.4%**.
- **Mastercard Code for Change 3.0: Finalist**.
- **Smart India Hackathon: Top 5** teams in the college-level round (2026); cleared the college-level selection round (2025).
- **Problem Solving:** Solved **200+ problems** across LeetCode, CodeChef, and GeeksforGeeks.